analogy is to humans who settle a dispute quickly and without fuss by tossing a coin.

The conditional strategy ‘If you are the resident, attack; if you are the intruder, retreat’ could be an ESS. Since the asymmetry is assumed to be arbitrary, the opposite strategy ‘If resident, retreat; if intruder, attack’ could also be stable. Which of the two ESSs is adopted in a particular population would depend on which one happens to reach a majority first. Once a majority of individuals is playing one of these two conditional strategies, deviants from it are penalized. Hence, by definition, it is an ESS.

For instance, suppose all individuals are playing ‘resident wins, intruder runs away’. This means they will win half their fights and lose half their fights. They will never be injured and they will never waste time, since all disputes are instantly settled by arbitrary convention. Now consider a new mutant rebel. Suppose he plays a pure hawk strategy, always attacking and never retreating. He will win when his opponent is an intruder. When his opponent is a resident he will run a grave risk of injury. On average he will have a lower pay-off than individuals playing according to the arbitrary rules of the ESS. A rebel who tries the reverse convention ‘if resident run away, if intruder attack’, will do even worse. Not only will he frequently be injured, he will also seldom win a contest. Suppose, though, that by some chance events individuals playing this reverse convention managed to become the majority. In this case their strategy would then become the stable norm, and deviation from *it* would be penalized. Conceivably, if we watched a population for many generations we would see a series of occasional flips from one stable state to the other.

However, in real life, truly arbitrary asymmetries probably do not exist. For instance, residents probably tend to have a practical advantage over intruders. They have better knowledge of local terrain. An intruder is perhaps more likely to be out of breath because he moved into the battle area, whereas the resident was there all the time. There is a more abstract reason why, of the two stable states, the ‘resident wins, intruder retreats’ one is the more probable in nature. This is that the reverse strategy, ‘intruder wins, resident retreats’ has an inherent tendency to self-destruction—it is what Maynard Smith would call a paradoxical strategy. In any population sitting at this paradoxical ESS, individuals would always be striving never to be caught as residents: they would